Speaker Script — Numbers That Talk: divisibility rules and the modular arithmetic behind them

Slide 1 — Title: Numbers That Talk (Zhiheng, about 30 s)

Hi everyone. I'm Zhiheng, and this is Yueran. Our project is about divisibility rules. You know the kind: "add up the digits to check for 3." Quick question. This year is 2026. **Is 2026 divisible by 9?** Don't reach for your phone. 2 plus 0 plus 2 plus 6 is 10. Not a multiple of 9, so no. And 10 leaves remainder 1. So does 2026. Everyone knows this trick. Almost nobody knows why it works. In ten minutes you will. But first, a magic trick.

Click to show the question. Give the class 5 seconds to add.

Slide 2 — Magic trick 1: the vanishing digit (Zhiheng, about 75 s)

Everyone, pick a four-digit number. Any number. Just don't use the same digit four times. Don't tell me. Now shuffle the digits. Make a new number out of the same digits. Subtract the smaller one from the bigger one. Got a result? Good. **Now circle one digit in your result. Not a zero.** Circle something that isn't zero. Hide it. I need one volunteer. Read me the other digits, in any order. Your hidden digit is... [answer]. Right? Let's do one more with someone else. I'm not reading minds. The number 9 is doing all the work. Let me show you.

Add the digits they read in your head. Answer = what is missing to reach the next multiple of 9. If the sum is already a multiple of 9, the answer is 9.

Example on the slide: 5274 → 7425 → 2151, hide 5, hear 2, 1, 1, sum 4, answer 5. Backup: 8317 → 1378 → 6939, hide a 9, hear 6, 3, 9, sum 18, answer 9. Faster 3-digit version: 831 → 138 → 693.

If the difference is 0 they shuffled into the same number. Ask them to shuffle again.

Slide 3 — A clock with only 9 marks (Zhiheng, about 75 s)

Here's the secret. Take 2026. It's really 2 thousands, 0 hundreds, 2 tens, and 6. **Now look at this clock. It only has 9 marks. Start at the top and walk 10 steps. You end up one mark past the start. Same as walking 1 step.** A hundred steps? Still one mark. A thousand? Still one. So on this clock, 2026 is just 2 plus 0 plus 2 plus 6. That's all "10 is congruent to 1 mod 9" means. Now the trick. Your number and your shuffled number have the same digits. So they land on the same mark. Subtract them, and you land on the top. That's zero. So the digits of your result add up to a multiple of 9. You read me all but one. I just fill in what's missing.

Point at the clock. Walk your finger 10 steps once, slowly.

Slide 4 — Bigger chunks: 99 and 999 (Zhiheng, about 75 s)

Why stop at single digits? 100 is 99 plus 1. So on a 99-clock, 100 acts like 1. That means: cut your number into two-digit chunks, add the chunks, and you get the remainder mod 99. And mod 11 and 33, for free. Now try 1000. That's 999 plus 1. And 999 is 27 times 37. So three-digit chunks test 27 and 37. **Here's one: 45658. Chunk it from the right: 45 and 658. Add them: 703. And 703 is 19 times 37. So 37 divides 45658.** No long division. We checked. It's 37 times 1234. One warning. Always chunk from the right. 45 | 658, not 456 | 58. The chunks are the "digits" when you write the number in base 1000.

Slide 5 — What about 7? (Zhiheng, about 45 s)

So the whole family of rules comes from one fact: some power of 10 acts like 1. 10 for 9. 100 for 99. 1000 for 999. **But look what's missing. 7. 13. 999 is 27 times 37. No 7, no 13.** Does 7 even have a rule? When we started, I thought no. Turns out there's a second family. In that family a power of 10 acts like minus 1. Yueran will show you. With a trick that needs three of your calculators.

Hand over. Yueran is already picking three people with phones out.

Slide 6 — Magic trick 2: 1001 brings it back (Yueran, about 75 s)

Thanks, Zhiheng. I need three people with calculators. You, you, and you. Everyone else, pick any three-digit number. Let's say 572. **Write it twice, so you get a six-digit number. 572572. First calculator: divide that by 7.** It divides evenly. I promise. **Second calculator: take that answer and divide by 11.** Also even. **Third: divide by 13.** What do you get? 572. Your original number. Everyone try it with your own number. It works every time. Why? Hold that thought.

Let each volunteer say their number out loud. Pause before the third one.

572 → 572572 → ÷7 = 81796 → ÷11 = 7436 → ÷13 = 572. Works with a zero: 205 → 205205 → 29315 → 2665 → 205. The trick cannot fail; a decimal on the calculator means a typo.

Slide 7 — Why it works: 1000 acts like –1 (Yueran, about 60 s)

Writing abc twice is abc times 1000, plus abc. That's abc times 1001. And 1001 is 7 times 11 times 13. So dividing by 7, then 11, then 13 is just dividing by 1001. You get abc back. Now the real point. 1000 is 1001 minus 1. On a 1001-clock, 1000 acts like minus 1. So three-digit chunks, with alternating signs, give the remainder mod 1001. And so mod 7, 11 and 13 at the same time. **Try 2026. Chunks 2 and 026. 026 minus 2 is 24.** 24 mod 7 is 3. 24 mod 13 is 11. Both correct. Same idea one digit at a time. 10 is 11 minus 1. **Alternate the digits, starting from the right: 6 minus 2 plus 0 minus 2 is 2.** And 2026 mod 11 is 2.

Slide 8 — One idea, every rule (Yueran, about 60 s)

So here's the one idea behind everything. **Find a k so that 10 to the k is 1 or minus 1 mod d. If it's 1, add the k-digit chunks. If it's minus 1, alternate them.** That's it. Here are a few. For 3 and 9, k is 1. For 11, 10 is already minus 1. For 37, 1000 is 1. For 7 and 13, 1000 is minus 1. For 101, 100 is minus 1. Every rule on this table is the same rule with a different k. Now the question we really wanted to answer. Does every d get a rule? Or is 7 just lucky?

Slide 9 — Why every prime gets a rule (Yueran, about 75 s)

Every prime other than 2 and 5 gets a rule. **That's Fermat's little theorem: 10 to the p minus 1 is 1 mod p.** So chunks of p minus 1 digits always work. Usually something shorter works. The shortest additive chunk is the order of 10 mod p. And here's what surprised us most. That same number is the length of the repeating block of 1 over p. **1/7 is 0.142857, repeating. Six digits. The order of 10 mod 7 is 6.** Same number, and it's not a coincidence. **Look at the remainders when you do the long division: 1, 3, 2, 6, 4, 5, and back to 1.** Those are the powers of 10 mod 7. One more. Multiply 142857 by 2: 285714. By 3: 428571. Same digits, rotated. That's because 10 is a primitive root mod 7. Our favourite fact in the whole project.

Trace the remainder cycle with your finger: 1 → 3 → 2 → 6 → 4 → 5.

Slide 10 — One sentence (Yueran, about 30 s)

One sentence. **If you know what 10 is mod d, you can build the rule for d yourself.** Fermat says the rule exists. The order of 10 says how short it can be. The full slides have more: a "chop the last digit" rule for 7, why ISBN numbers use 11, and every proof. It's all on our GitHub. Thanks. Go try the 1001 trick on a friend.

Likely questions (one or two sentences each)

Q1. Why don't 2 and 5 have a digit-sum rule?

Because 10 is 0 mod 2 and mod 5. 10 is not a unit there, so it has no order. But 10 being 0 is exactly why the last digit, or the last k digits, is enough.

Q2. Does every number have a rule?

Every d coprime to 10 does; Euler gives 10 to the phi of d is 1. A general d splits into a power of 2, a power of 5 and a part coprime to 10. Test each part on its own.

Q3. Is there a shorter rule for 7?

Yes. Chop the last digit y and compute x minus 2y, repeat. 2026 goes to 202 minus 12, that is 190, then 19, not a multiple of 7. It uses the inverse of 10 mod 7, which is 5, or minus 2. It only keeps yes or no, not the remainder.

Q4. Can I start the alternating sum from the left?

For yes or no, yes. For the remainder, no: with an even number of digits the sign flips. 2026 from the left gives minus 2, which is 9, but the remainder is 2.

Q5. Why does 1/7 repeat every 6 digits but 1/37 every 3?

The period of 1 over d is the order of 10 mod d. The order mod 7 is 6; mod 37 it is 3, because 999 is 27 times 37.

Q6. Can the tricks fail?

Trick A fails only if someone circles a 0, since 0 and 9 look the same, or shuffles into the same number. Trick B cannot fail.

Q7. How is this related to cryptography?

The order of an element and primitive roots are the core of Diffie-Hellman and ElGamal. Check digits, like ISBN-10 with mod 11, are divisibility rules used in practice.

Q8. Why chunk from the right?

Because n is B zero plus B one times 10 to the k plus B two times 10 to the 2k and so on. B zero has to be the lowest k digits.

• $99 = 9 \times 11$；$999 = 27 \times 37$；$1001 = 7 \times 11 \times 13$；$99999 = 9 \times 41 \times 271$。